

PERSONAL DETAILS

Rui Mo

Munich, Germany | ✉ rui.mo@tum.de | ☎ +49 176 4160 2949 | [ORCID](#) | [Google Scholar](#)

Research Interests: Electrochemical Biosensors, Functional Hydrogel Coatings, Protein/Biopolymer Self-Assembly, Bio-interfaces, Nanoscale Molecular Transport, and Sustainable Functional Materials.

EDUCATION

06/2026- Present	Massachusetts Institute of Technology (MIT)	USA
	Visiting student/Master's Thesis Researcher	
04/2025- Present	Technical University of Munich (TUM)	Germany
	Master of Science in <i>Technology of Biogenic Resources</i>	GPA: 2.0/1.0 (1.0 = highest)
09/2021 - 07/2023	China Agricultural University	China
	Master of Science in <i>Resources and Environmental Sciences</i>	GPA: 3.59/4.00
09/2017 - 06/2021	Changchun University of Technology	China
	Bachelor in <i>Environmental Engineering</i>	GPA: 3.30/4.00

RESEARCH EXPERIENCE

Broad Institute of MIT and Harvard & MIT (L4TE) USA

Master's Thesis Researcher & Supervisor: Prof. [Giovanni Traverso](#) and Dr. Isaac Tucker **06/2026-Present**

Internship

Development of Coatings for Implantable Gold-Based Electrochemical Aptamer Biosensors

- Developing hydrogel-based protective coatings for implantable gold electrodes to improve biocompatibility and long-term signal stability
- Optimizing coating composition, thickness, and interfacial adhesion to regulate molecular diffusion and minimize biofouling
- Evaluating biosensors performance through electrochemical characterization, including CV, EIS, under biologically relevant conditions

Technical University of Munich Germany

Internship & Research Supervisor: Prof. [Dr. Cordt Zollfrank](#) and Dr. Weixuan Wang **04/2025-05/2026**

Assistant

Clay-modified Cellulose Porous Foams for Thermal Insulation and Flame Resistance

- Synthesized sustainable, lightweight bio-based porous foams by blending native/regenerated cellulose fibers with functional clay minerals via optimized freeze-drying protocols.
- Enhanced cellulose-clay interfacial compatibility through organic surface modification of clay platelets, resolving phase separation issues and achieving highly ordered porous architectures.
- Regulated pore morphology, mechanical elasticity, thermal conductivity, and flame-retardant properties by systematically tuning the multi-component formulations.

China Agricultural University China

Research Assistant Supervisor: Prof. Changwei Zhao **07/2023-04/2024**

- Supported membrane fabrication, performance evaluation, data analysis, and laboratory coordination across multiple nanofiltration projects.

China Agricultural University China

Master's Research Supervisor: Prof. Changwei Zhao **09/2021-07/2023**

1. Polyamide Membranes with Nanoscale Ordered Structures for Fast Permeation and Highly Selective Ion-Ion Separation

- Regulated interfacial polymerization using g-C₃N₄ nanosheets to control polyamide network formation and nanoscale ordering
- Achieved ultra-high permeance (105 L·m⁻²·h⁻¹·bar⁻¹) with 99.4% Na₂SO₄ rejection and high Cl⁻/SO₄²⁻ selectivity (130)
- Investigated selective monomer–nanosheet interactions and transport mechanisms using molecular dynamics simulations
- Established structure–performance relationships between polyamide network ordering and ion separation behavior

2. Functionalized MoS₂ based polyamide composite NF membranes for enhanced separation performance

- Developed functionalized MoS₂ nanosheet-based thin-film nanocomposite membranes via interfacial polymerization
- Improved hydrophilicity, surface charge, permeability, and sulfate rejection through tailored nanosheet functionalization.
- Achieved high water flux (up to 58 L·m⁻²·h⁻¹·bar⁻¹) with sulfate rejection above 98% through MoS₂/HPE synergistic modification
- Investigated ion-transport behavior using experiments and molecular dynamics simulations.

Changchun University of Technology

China

Undergraduate Research

Supervisor: Prof. Xudong Yang

09/2020-06/2021

Biomass-based Carbonized Polymer Dots for Trace Ag⁺ Detection

- Synthesized carbonized polymer dots using Lycium barbarum via hydrothermal methods.
- Developed a rapid, low-toxicity fluorescence-based method for detecting trace Ag⁺ in environmental samples.
- Demonstrated low detection limits, good stability, and potential for portable detection devices.

PUBLICATIONS

Papers in SCI Journals:

1. Changwei Zhao*, Yanjun Zhang, Yuewen Jia, Bojun Li, Wenjing Tang, Chuning Shang, **Rui Mo**, Pei Li, Shaomin Liu & Sui Zhang*. Polyamide membranes with nanoscale ordered structures for fast permeation and highly selective ion-ion separation. *Nat Commun* **14** (1), 1112, 2023. (These authors contributed equally: Changwei Zhao, Yanjun Zhang, Yuewen Jia, Bojun Li.)
2. Junliang Huang*, Boyuan Xuan, Yanwen Ma, **Rui Mo**, Jiekai Wang, Changwei Zhao*. Enhanced boron rejection and permeability of thin-thin nanocomposite membranes via g-C₃N₄-tuned diffusion and reaction. *Desalination*, 118902, 2025.
3. Wenjing Tang, Bojun Li, Boyuan Xuan, Shaomin Liu, **Rui Mo**, Sihan Shao, Yushuang Liang, Yanwen Wei, Changwei Zhao. Interfacial construction of PIM-based interlayers in nanofiltration membranes for efficient saline wastewater treatment. 2025.

Papers in Chinese Journals:

1. Yinghai Zhao, Wenjing Tang, **Rui Mo**, Yiran Zhang, Changwei Zhao*, Rongfeng Jiang. Research on the Combined Treatment of Rural Domestic Wastewater Using Soil Infiltration Systems and Membrane Separation Systems. *Technology of Water Treatment*, 48(10): 127-130 (2022).
2. Zheming Shi, Lin Guo, Guiming Fang, Beibei Fan, **Rui Mo**, Changwei Zhao*. Advances in Microbial Remediation of Polycyclic Aromatic Hydrocarbon-Contaminated Soil. *Modern Chemical Industry*, 42(10): 24-28 (2022).

Patents

1. Changwei Zhao, **Rui Mo**. A preparation method of nanofiltration membrane with high permeability. CN202211136638.9.
2. Changwei Zhao, Wenjing Tang, **Rui Mo**, Yuqi Wang. A preparation method of oxidation-resistant and pollution-resistant polyester nanofiltration membrane. CN202210306696.5

PROFESSIONAL SKILLS

- Material Syntheses: Biomass-derived nanomaterials, 2D nanosheets (g-C₃N₄, MoS₂), interfacial polymerization, biopolymer formulation, and hydrogel network construction.
- Characterization Techniques: Proficient in FT-IR, SEM, AFM, TEM, XPS, XRD, and TGA.
- Electrochemical Analysis: Experienced in CV, EIS, and bio-interface sensing analysis under physiologically relevant settings.
- Molecular Simulations: Hands-on experience in Molecular Dynamics (MD) simulations using Materials Studio and LAMMPS to investigate molecular/ionic transport.

AWARDS & HONORS

- NUS Suzhou Research Institute Summer Programme Certificate, 2023
- First-Class Academic Scholarship, China Agricultural University, 2022
- Outstanding Graduation Thesis Award, Changchun University of Technology, 2021
- First-Class Academic Scholarship, Changchun University of Technology, 2019–2021